

Jinnah University For Women



Web Engineering

Project Proposal

Submitted To: Sir Saad

Group members

Ayesha Ahmed Siddiqui

Anzeela Siddiqi

Department of Computer Science & Software Engineering

Converso — Where Conversations Teach

1. Introduction

Communication is one of the most important parts of human interaction, yet language barriers still prevent people from connecting effectively across the world. While many tools exist for translation, they are often separate from real communication platforms or lack natural conversational flow.

Converso is designed to solve this problem by combining real-time chat, automatic translation, and language learning into a single platform. The goal is to allow users to communicate naturally, even if they do not share a common language.

2. Problem Statement

People from different countries often face difficulty communicating due to language differences. Existing solutions either:

- Require manual translation
- Are not integrated into chat systems
- Focus only on learning rather than real communication

As a result, users cannot experience smooth, natural conversations across languages.

3. Proposed Solution

Converso is a multilingual chat application that automatically translates messages between users in real time based on their preferred language.

Key idea:

Each user writes in their own language, and the system instantly converts the message into the receiver's language. Additionally, the platform supports language learning by showing both original and translated text, helping users improve their skills while chatting.

4. Objectives

The main objectives of this project are:

- To eliminate language barriers in communication
- To provide real-time automatic translation in chat
- To support language learning during conversations
- To create a simple, user-friendly, and modern chat interface

5. Key Features

- User authentication (email-based login and verification)
- Real-time messaging system
- Automatic translation of messages
- Multi-language support (initially 15–20 languages)
- Dual message display (original + translated text)
- Typing indicators and message status (sent, delivered, read)
- Theme customization (dark/light mode)
- Report and block system for safety
- Admin panel for monitoring and moderation

6. Methodology

The system will be developed using the MERN stack:

- Frontend: React
- Backend: Node.js and Express
- Database: MongoDB
- Real-time communication: Socket.IO

Translation will be implemented using APIs such as:

- Google Translate API
- DeepL API (for higher accuracy where available)

7. Expected Outcomes

- A fully functional multilingual chat platform
- Improved communication between users of different languages
- A user-friendly and modern interface

8. Conclusion

Converso aims to create a platform where people can communicate freely without worrying about language differences. By combining communication, translation, and learning, it provides a unique and practical solution for global interaction.